

BEAT Technical Report

AI for Good Innovation Challenge on Generative AI Applications for Enterprise Scenarios Using OPEA

OPEA-based Enterprise SOP Coaching and Assessment Agent

BEAT transforms enterprise SOP documents, training manuals, videos, audio transcripts, and structured files into a closed-loop training workflow: knowledge preparation, RAG-based Q&A, intelligent practice, answer assessment, evidence-backed feedback, and learning analytics.

Problem

- Frontline training relies on static manuals, fragmented mentoring, and repeated manual course creation.
- SOP updates are hard to distribute quickly across roles, departments, and sites.
- Generic Q&A lacks traceability, while scoring and follow-up coaching are difficult to control.

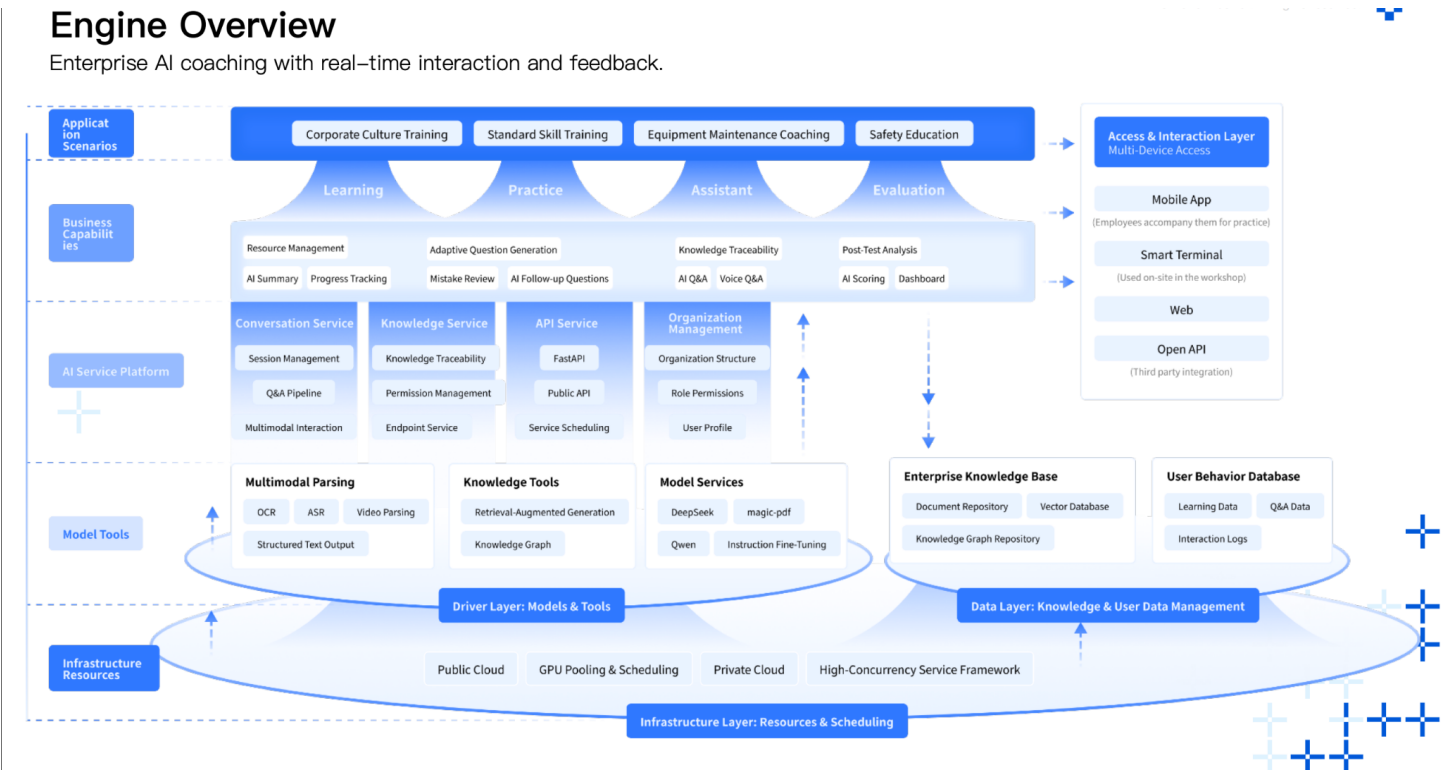
BEAT Solution

- Converts enterprise materials into traceable knowledge points, Q&A pairs, and practice questions.
- Uses RAG and vector retrieval to ground answers, scoring, and explanations in source content.
- Builds a learning-practice-assessment-feedback loop for SOP-driven workforce training.

AI for Good Value

- Makes workplace knowledge easier to access for frontline workers with fragmented time.
- Supports continuous skill improvement through short practice sessions and immediate feedback.
- Helps enterprises improve safety, onboarding, and procedure compliance without heavy training overhead.

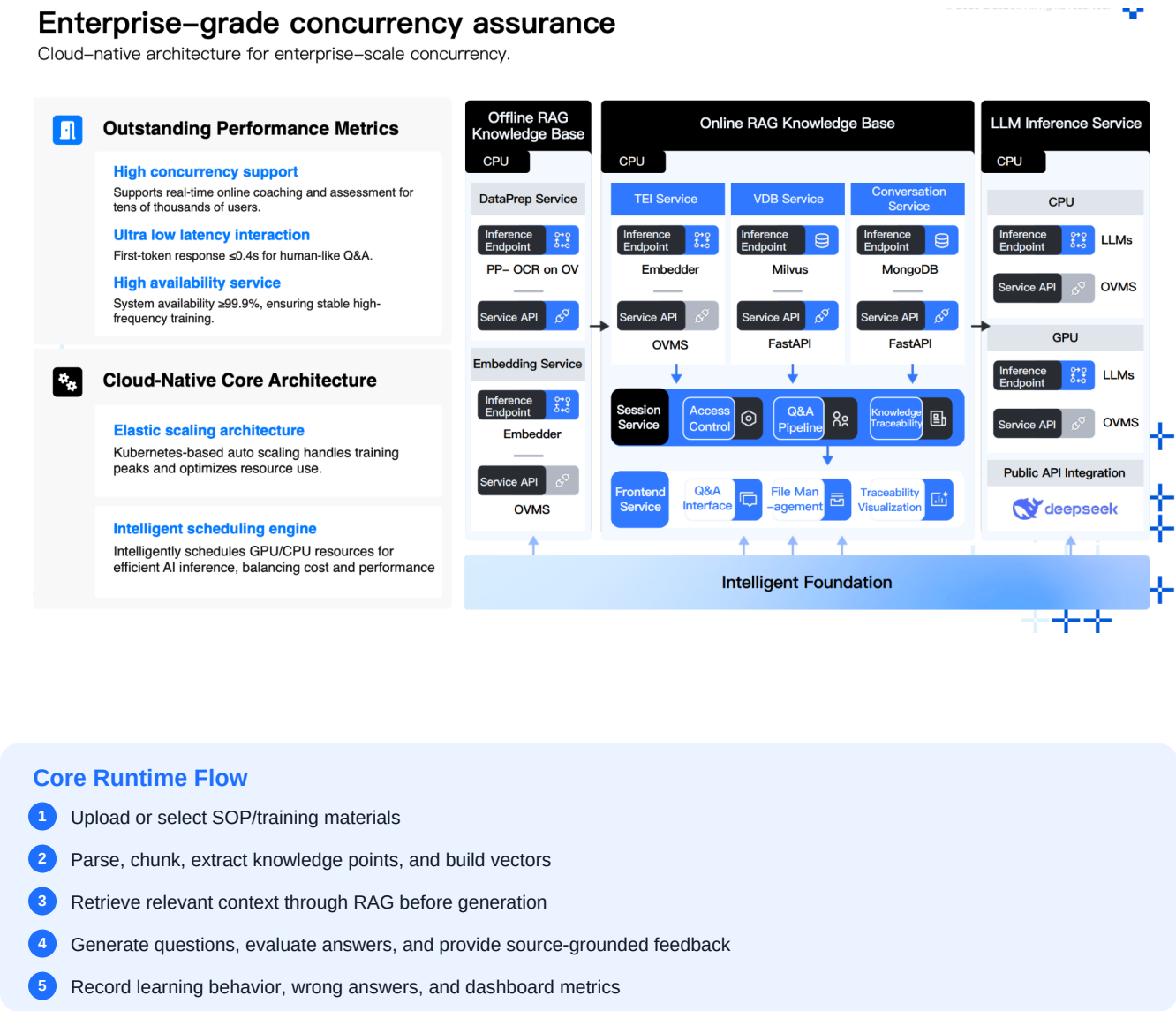
Figure 1. Engine Overview



Technical Architecture and Evaluation Readiness

Modular OPEA-style GenAI workflow for enterprise SOP training

Figure 2. Cloud-native Architecture



OPEA Component Mapping

- Knowledge ingestion: dataprep, ASR, Excel processing, document parsing.
- RAG layer: embeddings, Milvus vector database, retriever, rerank, and source-grounded context.
- LLM service: Q&A, question generation, answer evaluation, scoring, and explanation.
- Orchestration and guardrails: smart_practice workflow, session state, timer, answer logs, and traceability.

Deployment and Demo

- Single-node Docker Compose deployment is provided through deployment/docker-compose/quickstart-dfxw.sh.
- Local UI default: <https://localhost:5174>, configurable through EKBA_UI_PORT.
- Online evaluation demo: <https://14.103.144.187:30111/dashboard>, account user1-en / admin123.
- External model endpoints, API keys, ASR, and document parsing services are configured through .env.

Evaluation Focus

- Creativity and business value: procedure-anchored training for real enterprise frontline work.
- Technical implementation: modular microservices, RAG, vector database, LLM services, and orchestration.
- Prototype quality: end-to-end workflow from knowledge ingestion to exam coaching and feedback.
- Compliance: Apache-2.0 license, third-party notices, sanitized defaults, and no customer-sensitive data in public materials.